

OVP Guide to Using Processor Models

Model specific information for ARM_AArch64

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Chapter 1

Overview

This document provides the details of an OVP Fast Processor Model variant.

OVP Fast Processor Models are written in C and provide a C API for use in C based platforms. The models also provide a native interface for use in SystemC TLM2 platforms.

The models are written using the OVP VMI API that provides a Virtual Machine Interface that defines the behavior of the processor. The VMI API makes a clear line between model and simulator allowing very good optimization and world class high speed performance. Most models are provided as a binary shared object and also as source. This allows the download and use of the model binary or the use of the source to explore and modify the model.

The models are run through an extensive QA and regression testing process and most model families are validated using technology provided by the processor IP owners. There is a companion document (OVP Guide to Using Processor Models) which explains the general concepts of OVP Fast Processor Models and their use. It is downloadable from the OVPworld website documentation pages.

1.1 Description

ARM Processor Model

1.2 Licensing

Usage of binary model under license governing simulator usage.

Note that for models of ARM CPUs the license includes the following terms:

Licensee is granted a non-exclusive, worldwide, non-transferable, revocable licence to:

If no source is being provided to the Licensee: use and copy only (no modifications rights are granted) the model for the sole purpose of designing, developing, analyzing, debugging, testing, verifying, validating and optimizing software which: (a) (i) is for ARM based systems; and (ii) does not incorporate the ARM Models or any part thereof; and (b) such ARM Models may not be used

to emulate an ARM based system to run application software in a production or live environment.

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In the case of any Licensee who is either or both an academic or educational institution the purposes shall be limited to internal use.

Except to the extent that such activity is permitted by applicable law, Licensee shall not reverse engineer, decompile, or disassemble this model. If this model was provided to Licensee in Europe, Licensee shall not reverse engineer, decompile or disassemble the Model for the purposes of error correction.

The License agreement does not entitle Licensee to manufacture in silicon any product based on this model.

The License agreement does not entitle Licensee to use this model for evaluating the validity of any ARM patent.

Source of model available under separate Imperas Software License Agreement.

1.3 Limitations

Instruction pipelines are not modeled in any way. All instructions are assumed to complete immediately. This means that instruction barrier instructions (e.g. ISB, CP15ISB) are treated as NOPs, with the exception of any undefined instruction behavior, which is modeled. The model does not implement speculative fetch behavior. The branch cache is not modeled.

Caches and write buffers are not modeled in any way. All loads, fetches and stores complete immediately and in order, and are fully synchronous (as if the memory was of Strongly Ordered or Device-nGnRnE type). Data barrier instructions (e.g. DSB, CP15DSB) are treated as NOPs, with the exception of any undefined instruction behavior, which is modeled. Cache manipulation instructions are implemented as NOPs, with the exception of any undefined instruction behavior, which is modeled.

Real-world timing effects are not modeled: all instructions are assumed to complete in a single cycle.

Performance Monitors are implemented as a register interface only except for the cycle counter, which is implemented assuming one instruction per cycle.

TLBs are architecturally-accurate but not device accurate. This means that all TLB maintenance and address translation operations are fully implemented but the cache is larger than in the real device.

Debug registers are implemented but non-functional (which is sufficient to allow operating systems such as Linux to boot). Debug state is not implemented.

1.4 Verification

Models have been extensively tested by Imperas. ARM Cortex-A models have been successfully used by customers to simulate SMP Linux, Ubuntu Desktop, VxWorks and ThreadX on Xilinx Zynq virtual platforms.

1.5 Features

The precise set of implemented features in the model is defined by ID registers. Use overrides to modify these if required (for example `override_PFR0` or `override_AA64PFR0_EL1`).

1.5.1 Core Features

AArch64 is implemented at EL3, EL2, EL1 and EL0.

The following ARMv8.2 core features are implemented: FEAT_FP16, FEAT_SVE.

1.5.2 Memory System

Security extensions are implemented (also known as TrustZone). To make non-secure accesses visible externally, override `ID_AA64MMFR0_EL1.PARange` to specify the required physical bus size (32, 36, 40, 42, 44, 48 or 52 bits) and connect the processor to a bus one bit wider (33, 37, 41, 43, 45, 49 or 53 bits, respectively). The extra most-significant bit is the NS bit, indicating a non-secure access. If non-secure accesses are not required to be made visible externally, connect the processor to a bus of exactly the size implied by `ID_AA64MMFR0_EL1.PARange`.

VMSA EL1, EL2 and EL3 stage 1 address translation is implemented. VMSA stage 2 address translation is implemented.

LPA (large physical address extension) is implemented as standard in ARMv8.

TLB behavior is controlled by parameter `ASIDCacheSize`. If this parameter is 0, then an unlimited number of TLB entries will be maintained concurrently. If this parameter is non-zero, then only TLB entries for up to `ASIDCacheSize` different ASIDs will be maintained concurrently initially; as new ASIDs are used, TLB entries for less-recently used ASIDs are deleted, which improves model performance in some cases (especially when 16-bit ASIDs are in use). If the model detects that the TLB entry cache is too small (entry ejections are very frequent), it will increase the cache size automatically. In this variant, `ASIDCacheSize` is 8

1.5.3 Advanced SIMD and Floating-Point Features

SIMD and VFP instructions are implemented.

The model implements trapped exceptions if `FPTrap` is set to 1 in `MVFR0` (for AArch32) or `MVFR0_EL1` (for AArch64). When floating point exception traps are taken, cumulative exception flags are not updated (in other words, cumulative flag state is always the same as prior to instruction

execution, even for SIMD instructions). When multiple enabled exceptions are raised by a single floating point operation, the exception reported is the one in least-significant bit position in FPSCR (for AArch32) or FPCR (for AArch64). When multiple enabled exceptions are raised by different SIMD element computations, the exception reported is selected from the lowest-index-number SIMD operation. Contact Imperas if requirements for exception reporting differ from these.

Trapped exceptions are implemented in this variant (FPTrap=1)

1.5.4 SVE Features

SVE is implemented, with 512 bit vectors. The exact set of supported vector widths can be specified using parameter “SVEImplementedSizes”. This parameter is a mask in which vector size Nx128 is supported if bit $1 < (N-1)$ is set in the mask. For example, a mask value of 0xf indicates that 128, 256, 384 and 512 bit vector widths are supported. As a second example, a value of 0x8b indicates that 128, 256, 512 and 1024 bit vector widths are supported. In this variant, “SVEImplementedSizes” is 0xf.

For SVE First-fault and Non-fault vector loads, the UNKNOWN value returned for faulting loads (or those disabled by the FFR) can be specified using parameter “SVEFaultUnknown”. In this variant, “SVEFaultUnknown” is 0xdfdfdfdfdfdfdfdf.

When operating with a vector length less than the maximum length, the value of inaccessible vector and predicate elements is preserved.

1.5.5 Generic Timer

Generic Timer is present. Use parameter “override_timerScaleFactor” to specify the counter rate as a fraction of the processor MIPS rate (e.g. 10 implies Generic Timer counters increment once every 10 processor instructions).

1.6 Debug Mask

It is possible to enable model debug features in various categories. This can be done statically using the “override_debugMask” parameter, or dynamically using the “debugflags” command. Enabled debug features are specified using a bitmask value, as follows:

Value 0x004: enable debugging of MMU/MPU mappings.

Value 0x080: enable debugging of all system register accesses.

Value 0x100: enable debugging of all traps of system register accesses.

Value 0x200: enable verbose debugging of other miscellaneous behavior (for example, the reason why a particular instruction is undefined).

Value 0x400: enable debugging of Performance Monitor timers

Value 0x800: enable dynamic validation of TLB entries against in-memory page table contents (finds some classes of error where page table entries are updated without a subsequent flush of

affected TLB entries).

All other bits in the debug bitmask are reserved and must not be set to non-zero values.

1.7 Integration Support

This model implements a number of non-architectural pseudo-registers and other features to facilitate integration.

1.7.1 Memory Transaction Query

Three registers are intended for use within memory callback functions to provide additional information about the current memory access. Register `transactPL` indicates the processor execution level of the current access (0-3). Note that for load/store translate instructions (e.g. `LDRT`, `STRT`) the reported execution level will be 0, indicating an EL0 access. Register `transactAT` indicates the type of memory access: 0 for a normal read or write; and 1 for a physical access resulting from a page table walk. Register `atomicType` indicates whether the current instruction has any special atomic constraints (0 indicates no constraint, 1 indicates atomic, 2 indicates exclusive access).

1.7.2 Page Table Walk Query

A banked set of registers provides information about the most recently completed page table walk. There are up to six banks of registers: bank 0 is for stage 1 walks, bank 1 is for stage 2 walks, and banks 2-5 are for stage 2 walks initiated by stage 1 level 0-3 entry lookups, respectively. Banks 1-5 are present only for processors with virtualization extensions. The currently active bank can be set using register `PTWBankSelect`. Register `PTWBankValid` is a bitmask indicating which banks contain valid data: for example, the value 0xb indicates that banks 0, 1 and 3 contain valid data.

Within each bank, there are registers that record addresses and values read during that page table walk. Register `PTWBase` records the table base address, register `PTWInput` contains the input address that starts a walk, register `PTWOutput` contains the result address and register `PTWPgSize` contains the page size (`PTWOutput` and `PTWPgSize` are valid only if the page table walk completes). Registers `PTWAddressL0`-`PTWAddressL3` record the addresses of level 0 to level 3 entries read, respectively. Register `PTWAddressValid` is a bitmask indicating which address registers contain valid data: bits 0-3 indicate `PTWAddressL0`-`PTWAddressL3`, respectively, bit 4 indicates `PTWBase`, bit 5 indicates `PTWInput`, bit 6 indicates both `PTWOutput` and `PTWPgSize`. For example, the value 0x73 indicates that `PTWBase`, `PTWInput`, `PTWOutput`, `PTWPgSize` and `PTWAddressL0`-`L1` are valid but `PTWAddressL2`-`L3` are not. Register `PTWAddressNS` is a bitmask indicating whether an address is in non-secure memory: bits 0-3 indicate `PTWAddressL0`-`PTWAddressL3`, respectively, bit 4 indicates `PTWBase`, bit 6 indicates `PTWOutput` (`PTWInput` is a VA and thus has no secure/non-secure info). Registers `PTWValueL0`-`PTWValueL3` contain page table entry values read at level 0 to level 3. Register `PTWValueValid` is a bitmask indicating which value registers contain valid data: bits 0-3 indicate `PTWValueL0`-`PTWValueL3`, respectively.

1.7.3 Artifact Page Table Walks

Registers are also available to enable a simulation environment to initiate an artifact page table walk (for example, to determine the ultimate PA corresponding to a given VA). Register PTWI_EL1S initiates a secure EL1 table walk for a fetch. Register PTWD_EL1S initiates a secure EL1 table walk for a load or store (note that current ARM processors have unified TLBs, so these registers are synonymous). Registers PTW[ID]_EL1NS initiate walks for non-secure EL1 accesses. Registers PTW[ID]_EL2 initiate EL2 walks. Registers PTW[ID]_S2 initiate stage 2 walks. Registers PTW[ID]_EL3 initiate AArch64 EL3 walks. Finally, registers PTW[ID]_current initiate current-mode walks (useful in a memory callback context). Each walk fills the query registers described above.

1.7.4 MMU and Page Table Walk Events

Two events are available that allow a simulation environment to be notified on MMU and page table walk actions. Event mmuEnable triggers when any MMU is enabled or disabled. Event pageTableWalk triggers on completion of any page table walk (including artifact walks).

1.7.5 Artifact Address Translations

A simulation environment can trigger an artifact address translation operation by writing to the architectural address translation registers (e.g. ATS1CPR). The results of such translations are written to an integration support register artifactPAR, instead of the architectural PAR register. This means that such artifact writes will not perturb architectural state.

1.7.6 TLB Invalidation

A simulation environment can cause TLB state for one or more address translation regimes in the processor to be flushed by writing to the artifact register ResetTLBs. The argument is a bitmask value, in which non-zero bits select the TLBs to be flushed, as follows:

Bit 0: EL0/EL1 stage 1 secure TLB

Bit 1: EL0/EL1 stage 1 non-secure TLB

Bit 2: EL2 stage 1 non-secure TLB

Bit 3: EL0/EL1 stage 2 non-secure TLB

Bit 4: EL3 stage 1 TLB

1.7.7 Halt Reason Introspection

An artifact register HaltReason can be read to determine the reason or reasons that a processor is halted. This register is a bitfield, with the following encoding: bit 0 indicates the processor has executed a wait-for-event (WFE) instruction; bit 1 indicates the processor has executed a wait-for-interrupt (WFI) instruction; and bit 2 indicates the processor is held in reset.

1.7.8 Automatic WFI Wake Up

A simulation environment can cause a core to wake up from the WFI state by writing to the artifact register WakeWFINow.

1.7.9 System Register Access Monitor

If parameter “enableSystemMonitorBus” is True, an artifact 32-bit bus “SystemMonitor” is enabled for each PE. Every system register read or write by that PE is then visible as a read or write on this artifact bus, and can therefore be monitored using callbacks installed in the client environment (use opBusReadMonitorAdd/opBusWriteMonitorAdd or icmAddBusReadCallback-/icmAddBusWriteCallback, depending on the client API). The format of the address on the bus is as follows:

bits 31:26 - zero

bit 25 - 1 if AArch64 access, 0 if AArch32 access

bit 24 - 1 if non-secure access, 0 if secure access

bits 23:20 - CRm value

bits 19:16 - CRn value

bits 15:12 - op2 value

bits 11:8 - op1 value

bits 7:4 - op0 value (AArch64) or coprocessor number (AArch32)

bits 3:0 - zero

As an example, to view non-secure writes to writes to CNTFRQ_EL0 in AArch64 state, install a write monitor on address range 0x020e0330:0x020e0333.

1.7.10 System Register Implementation

If parameter “enableSystemBus” is True, an artifact 32-bit bus “System” is enabled for each PE. Slave callbacks installed on this bus can be used to implement modified system register behavior (use opBusSlaveNew or icmMapExternalMemory, depending on the client API). The format of the address on the bus is the same as for the system monitor bus, described above.

Chapter 2

Configuration

2.1 Location

This model's VLVN is `arm.ovpworld.org/processor/arm/1.0`.

The model source is usually at:

`$IMPERAS_HOME/ImperasLib/source/arm.ovpworld.org/processor/arm/1.0`

The model binary is usually at:

`$IMPERAS_HOME/lib/$IMPERAS_ARCH/ImperasLib/arm.ovpworld.org/processor/arm/1.0`

2.2 GDB Path

The default GDB for this model is: `$IMPERAS_HOME/lib/$IMPERAS_ARCH/gdb/aarch64-none-elf-gdb`.

2.3 Semi-Host Library

The default semi-host library file is `arm.ovpworld.org/semihosting/armAngel/1.0`

2.4 Processor Endian-ness

This is a LITTLE endian model.

2.5 QuantumLeap Support

This processor is qualified to run in a QuantumLeap enabled simulator.

2.6 Processor ELF code

The ELF code supported by this model is: `0xb7`.

Chapter 3

All Variants in this model

This model has these variants

Variant	Description
ARMv4T	
ARMv4xM	
ARMv4	
ARMv4TxM	
ARMv5xM	
ARMv5	
ARMv5TxM	
ARMv5T	
ARMv5TExP	
ARMv5TE	
ARMv5TEJ	
ARMv6	
ARMv6K	
ARMv6T2	
ARMv6KZ	
ARMv7	
ARM7TDMI	
ARM7EJ-S	
ARM720T	
ARM920T	
ARM922T	
ARM926EJ-S	
ARM940T	
ARM946E	
ARM966E	
ARM968E-S	
ARM1020E	
ARM1022E	
ARM1026EJ-S	
ARM1136J-S	
ARM1156T2-S	

ARM1176JZ-S	
Cortex-R4	
Cortex-R4F	
Cortex-R5MPx1	
Cortex-R5MPx2	
Cortex-R8MPx1	
Cortex-R8MPx2	
Cortex-R8MPx3	
Cortex-R8MPx4	
Cortex-R52MPx1	
Cortex-R52MPx2	
Cortex-R52MPx3	
Cortex-R52MPx4	
Cortex-R52+MPx1	
Cortex-R52+MPx2	
Cortex-R52+MPx3	
Cortex-R52+MPx4	
Cortex-R82MPx1	
Cortex-R82MPx2	
Cortex-R82MPx3	
Cortex-R82MPx4	
Cortex-R82MPx5	
Cortex-R82MPx6	
Cortex-R82MPx7	
Cortex-R82MPx8	
Cortex-A5UP	
Cortex-A5MPx1	
Cortex-A5MPx2	
Cortex-A5MPx3	
Cortex-A5MPx4	
Cortex-A8	
Cortex-A9UP	
Cortex-A9MPx1	
Cortex-A9MPx2	
Cortex-A9MPx3	
Cortex-A9MPx4	
Cortex-A7UP	
Cortex-A7MPx1	
Cortex-A7MPx2	
Cortex-A7MPx3	
Cortex-A7MPx4	
Cortex-A15UP	
Cortex-A15MPx1	
Cortex-A15MPx2	
Cortex-A15MPx3	

Cortex-A15MPx4	
Cortex-A17MPx1	
Cortex-A17MPx2	
Cortex-A17MPx3	
Cortex-A17MPx4	
AArch32	
AArch64	(described in this document)
Cortex-A32MPx1	
Cortex-A32MPx2	
Cortex-A32MPx3	
Cortex-A32MPx4	
Cortex-A35MPx1	
Cortex-A35MPx2	
Cortex-A35MPx3	
Cortex-A35MPx4	
Cortex-A53MPx1	
Cortex-A53MPx2	
Cortex-A53MPx3	
Cortex-A53MPx4	
Cortex-A55MPx1	
Cortex-A55MPx2	
Cortex-A55MPx3	
Cortex-A55MPx4	
Cortex-A57MPx1	
Cortex-A57MPx2	
Cortex-A57MPx3	
Cortex-A57MPx4	
Cortex-A65AEMPx1	
Cortex-A65AEMPx2	
Cortex-A65AEMPx3	
Cortex-A65AEMPx4	
Cortex-A72MPx1	
Cortex-A72MPx2	
Cortex-A72MPx3	
Cortex-A72MPx4	
Cortex-A73MPx1	
Cortex-A73MPx2	
Cortex-A73MPx3	
Cortex-A73MPx4	
Cortex-A75MPx1	
Cortex-A75MPx2	
Cortex-A75MPx3	
Cortex-A75MPx4	
Cortex-A76MPx1	
Cortex-A76MPx2	

Cortex-A76MPx3	
Cortex-A76MPx4	
Cortex-A78MPx1	
Cortex-A78MPx2	
Cortex-A78MPx3	
Cortex-A78MPx4	
MultiCluster	

Table 3.1: All Variants in this model

Chapter 4

Bus Master Ports

This model has these bus master ports.

Name	min	max	Connect?	Description
INSTRUCTION	32	53	mandatory	
DATA	32	53	optional	

Table 4.1: Bus Master Ports

Chapter 5

Bus Slave Ports

This model has no bus slave ports.

Chapter 6

Net Ports

This model has these net ports.

Name	Type	Connect?	Description
EVENTI	input	optional	Event input signal, active on rising edge
EVENTO	output	optional	Event output signal, active on rising edge
CNTVIRQ	output	optional	EL1 Virtual timer event (active high)
CNTPSIRQ	output	optional	EL3 Physical timer event (active high)
CNTPNSIRQ	output	optional	EL1 Physical timer event (active high)
CNTPHIRQ	output	optional	Non-secure EL2 Physical timer event (active high)
CLUSTERIDAFF1	input	optional	Configure MPIDR.Aff1
CLUSTERIDAFF2	input	optional	Configure MPIDR.Aff2
CLUSTERIDAFF3	input	optional	Configure MPIDR.Aff3
RVBARADDRx	input	optional	Configure AArch64 Reset Vector Base Address at reset
VINITHI	input	optional	Configure HIVECS mode (SCTLR.V)
CFGEND	input	optional	Configure exception endianness (SCTLR.EE)
CFGTE	input	optional	Configure exception state at reset (SCTLR.TE)
reset	input	optional	Processor reset, active high
fiq	input	optional	FIQ interrupt, active high (negation of nFIQ)
irq	input	optional	IRQ interrupt, active high (negation of nIRQ)
sei	input	optional	System error interrupt, active on rising edge (negation of nSEI)
vfiq	input	optional	Virtual FIQ interrupt, active high (negation of nVFIQ)
virq	input	optional	Virtual IRQ interrupt, active high (negation of nVIRQ)
vsei	input	optional	Virtual system error interrupt, active on rising edge (negation of nVSEI)
haltReason	output	optional	Indicates why core is halted

AXI.SLVERR	input	optional	AXI external abort type (DECERR=0, SLVERR=1)
CP15SDISABLE	input	optional	CP15SDISABLE (active high)
PMUIRQ	output	optional	Performance monitor event (active high)
syncCluster	output	optional	Cluster synchronization required
updateTimer	output	optional	Internal timer update

Table 6.1: Net Ports

Chapter 7

FIFO Ports

This model has no FIFO ports.

Chapter 8

Formal Parameters

Name	Type	Description
verbose	Boolean	Specify verbosity of output
suppressCPSWarnings	Boolean	Suppress duplicate warnings generated using ARM_CP_CPSI or ARM_CP_CPSD message identifiers
showHiddenRegs	Boolean	Show hidden registers during register tracing
UAL	Boolean	Disassemble using UAL syntax
enableVFPAAtReset	Boolean	Enable vector floating point (SIMD and VFP) instructions at reset. (Enables cp10/11 in CPACR and sets FPEXC.EN)
SVEImplementedSizes	Uns32	For processors with ARMv8.2 SVE extension, mask of configurable vector sizes (vector length N is configurable if mask contains $1 < ((N/128)-1)$)
SVEFaultUnknown	Uns64	For processors with ARMv8.2 SVE extension, UNKNOWN value returned for suppressed or inactive FFR elements
enableSystemBus	Boolean	Add 32-bit artifact System bus port, allowing system registers to be externally implemented
enableSystemMonitorBus	Boolean	Add 32-bit artifact SystemMonitor bus port, allowing system register accesses to be externally monitored
compatibility	Enumeration	Specify compatibility mode
	ISA	
	gdb	
	nopSVC	
unpredictableR15	Enumeration	Specify behavior for UNPREDICTABLE uses of AArch32 R15 register
	undefined	
	nop	
	raz_wi	
	execute	
	assert	
unpredictableModal	Enumeration	Specify behavior for UNPREDICTABLE instructions in certain AArch32 modes (for example, MRS using SPSR in System mode)
	undefined	
	nop	
	assert	

maxSIMDUnroll	Uns32	If SIMD operations are supported, specify the maximum number of parallel SIMD operations to unroll (unrolled operations can be faster, but produce more verbose JIT code)
override.debugMask	Uns32	Specifies debug mask, enabling debug output for model components
ASIDCacheSize	Uns32	Specifies the number of different ASIDs for which TLB entries are cached; a value of 0 implies no limit
thumbNoCond	Boolean	Specify whether trapped Thumb instructions set CV=1 and COND field in syndrome (if False, both are zero)
enableHostAtomics	Boolean	Enable use of host atomic instructions to emulate load/store exclusive operations (not architecturally accurate, accelerates parallel simulation)
enableWFEHostAtomics	Boolean	Enable WFE and SEV instructions when using host atomics (does nothing otherwise)
override.fcsePresent	Boolean	Specifies that FCSE is present (if true)
override.fpexcDexPresent	Boolean	Specifies that the FPEXC.DEX register field is implemented (if true)
override.advSIMDPresent	Boolean	Specifies that Advanced SIMD extensions are present (if true)
override.vfpPresent	Boolean	Specifies that VFP extensions are present (if true)
override.physicalBits	Uns32	Specifies the implemented physical bus bits (defaults to connected physical bus width)
allowMSRToCurrentBank	Boolean	Allow access to the current mode registers via the MSR/MRS (Banked Register) instructions
override.SCTLR_V	Boolean	Override SCTLR.V with the passed value (enables high vectors; also configurable using VINITHI pin)
override.SCTLR_EE	Boolean	Override SCTLR.EE with the passed value (configures exception data endianness; also configurable using CFGEE pin)
override.SCTLR_TE	Boolean	Override SCTLR.TE with the passed value (configures Thumb state for exception handling; also configurable using TEINIT pin)
override.SCTLR_NMFI	Boolean	Override SCTLR.NMFI with the passed value (configures NMFI state for exception handling; also configurable using CFGNMFI pin)
override.SCTLR_CP15BEN_Present	Boolean	Enable ARMv7 SCTLR.CP15BEN bit (CP15 barrier enable)
override.MIDR	Uns32	Override MIDR/MIDR_EL1 register
override.CTR	Uns32	Override CTR/CTR_EL0 register
override.TLBTR	Uns32	Override TLBTR register
override.CLIDR	Uns32	Override CLIDR/CLIDR_EL1 register
override.AIDR	Uns32	Override AIDR/AIDR_EL1 register
override.PFR0	Uns32	Override ID_PFR0/ID_PFR0_EL1 register
override.PFR1	Uns32	Override ID_PFR1/ID_PFR1_EL1 register
override.PFR2	Uns32	Override ID_PFR2/ID_PFR2_EL1 register
override.DFR0	Uns32	Override ID_DFR0/ID_DFR0_EL1 register
override.DFR1	Uns32	Override ID_DFR1/ID_DFR1_EL1 register
override.AFR0	Uns32	Override ID_AFR0/ID_AFR0_EL1 register
override.MMFR0	Uns32	Override ID_MMFR0/ID_MMFR0_EL1 register

override_MMFR1	Uns32	Override ID_MMFR1/ID_MMFR1_EL1 register
override_MMFR2	Uns32	Override ID_MMFR2/ID_MMFR2_EL1 register
override_MMFR3	Uns32	Override ID_MMFR3/ID_MMFR3_EL1 register
override_MMFR4	Uns32	Override ID_MMFR4/ID_MMFR4_EL1 register
override_MMFR5	Uns32	Override ID_MMFR5/ID_MMFR5_EL1 register
override_ISAR0	Uns32	Override ID_ISAR0/ID_ISAR0_EL1 register
override_ISAR1	Uns32	Override ID_ISAR1/ID_ISAR1_EL1 register
override_ISAR2	Uns32	Override ID_ISAR2/ID_ISAR2_EL1 register
override_ISAR3	Uns32	Override ID_ISAR3/ID_ISAR3_EL1 register
override_ISAR4	Uns32	Override ID_ISAR4/ID_ISAR4_EL1 register
override_ISAR5	Uns32	Override ID_ISAR5/ID_ISAR5_EL1 register
override_ISAR6	Uns32	Override ID_ISAR6/ID_ISAR6_EL1 register
override_PMCr	Uns32	Override PMCr/PMCr_EL0 register (not functionally significant in the model)
override_PMCEID0	Uns64	Override PMCEID0/PMCEID0_EL0 register (not functionally significant in the model)
override_PMCEID1	Uns64	Override PMCEID1/PMCEID1_EL0 register (not functionally significant in the model)
override_PMMIR	Uns32	Override PMMIR/PMMIR_EL1 register (not functionally significant in the model)
override_DBGDIDR	Uns32	Override DBGDIDR register (not functionally significant in the model)
override_DBGDEVID	Uns32	Override DBGDEVID register (not functionally significant in the model)
override_DBGDEVID1	Uns32	Override DBGDEVID1 register (not functionally significant in the model)
override_DBGDEVID2	Uns32	Override DBGDEVID2 register (not functionally significant in the model)
override_FPSID	Uns32	Override SIMD/VFP FPSID register
override_MVFR0	Uns32	Override SIMD/VFP MVFR0/MVFR0_EL1 register
override_MVFR1	Uns32	Override SIMD/VFP MVFR1/MVFR1_EL1 register
override_MVFR2	Uns32	Override SIMD/VFP MVFR2/MVFR2_EL1 register
override_FPEXC	Uns32	Override SIMD/VFP FPEXC/FPEXC32_EL2 register
override_ERG	Uns32	Specifies exclusive reservation granule
override_CCSIDR_1I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 1 instruction)
override_CCSIDR_1D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 1 data)
override_CCSIDR_2I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 2 instruction)
override_CCSIDR_2D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 2 data)
override_CCSIDR_3I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 3 instruction)
override_CCSIDR_3D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 3 data)

override_CCSIDR_4I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 4 instruction)
override_CCSIDR_4D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 4 data)
override_CCSIDR_5I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 5 instruction)
override_CCSIDR_5D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 5 data)
override_CCSIDR_6I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 6 instruction)
override_CCSIDR_6D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 6 data)
override_CCSIDR_7I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 7 instruction)
override_CCSIDR_7D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 7 data)
override_RMR	Uns32	Override RMR register alias at highest-implemented exception level
override_RVBAR	Uns64	Override RVBAR register alias at highest-implemented exception level
override_AA64PFR0_EL1	Uns64	Override ID_AA64PFR0_EL1 register
override_AA64PFR1_EL1	Uns64	Override ID_AA64PFR1_EL1 register
override_AA64DFR0_EL1	Uns64	Override ID_AA64DFR0_EL1 register
override_AA64DFR1_EL1	Uns64	Override ID_AA64DFR1_EL1 register
override_AA64AFR0_EL1	Uns64	Override ID_AA64AFR0_EL1 register
override_AA64AFR1_EL1	Uns64	Override ID_AA64AFR1_EL1 register
override_AA64ISAR0_EL1	Uns64	Override ID_AA64ISAR0_EL1 register
override_AA64ISAR1_EL1	Uns64	Override ID_AA64ISAR1_EL1 register
override_AA64ISAR2_EL1	Uns64	Override ID_AA64ISAR2_EL1 register
override_AA64MMFR0_EL1	Uns64	Override ID_AA64MMFR0_EL1 register
override_AA64MMFR1_EL1	Uns64	Override ID_AA64MMFR1_EL1 register
override_AA64MMFR2_EL1	Uns64	Override ID_AA64MMFR2_EL1 register
override_DCZID_EL0	Uns32	Override DCZID_EL0 register
override_LORID_EL1	Uns32	Override LORID_EL1 register (ARMv8.1 only)
override_STRoffsetPC12	Boolean	Specifies that STR/STR of PC should do so with 12:byte offset from the current instruction (if true), otherwise an 8:byte offset is used
override_fcseRequiresMMU	Boolean	Specifies that FCSE is active only when MMU is enabled (if true)
override_ignoreBadCp15	Boolean	Specifies whether invalid coprocessor 15 access should be ignored (if true) or cause Invalid Instruction exceptions (if false)
override_SGIDisable	Boolean	Override whether GIC SGIs may be disabled (if true) or are permanently enabled (if false)
override_condUndefined	Boolean	Force undefined instructions to take Undefined Instruction exception even if they are conditional
override_deviceStrongAligned	Boolean	Force accesses to Device and Strongly Ordered regions to be aligned
override_stage1SZMinFault	Boolean	Enable Level 0 Translation faults when stage 1 TCR_ELx.TxSZ < minimum (by default, clamp to minimum)

override_stage1SZMaxFault	Boolean	Enable Level 0 Translation faults when stage 1 TCR_ELx.TxSZ >maximum (by default, clamp to maximum)
override_stage2SZMinFault	Boolean	Enable Level 0 Translation faults when stage 2 VTCR_EL2.T0SZ <minimum (by default, clamp to minimum)
override_stage2SZMaxFault	Boolean	Enable Level 0 Translation faults when stage 2 VTCR_EL2.T0SZ >maximum (by default, clamp to maximum)
override_mask_ACTLR_EL1	Uns64	Override mask of writable bits in AArch64 ACTLR_EL1 register, or AArch32 non-secure ACTLR/ACTLR2 pair, if implemented
override_mask_ACTLR_EL2	Uns64	Override mask of writable bits in AArch64 ACTLR_EL2 register, or AArch32 HACTLR/HACTLR2 pair, if implemented
override_mask_ACTLR_EL3	Uns64	Override mask of writable bits in AArch64 ACTLR_EL3 register, or AArch32 secure ACTLR/ACTLR2 pair, if implemented
override_Control_V	Boolean	Override SCTLR.V with the passed value (deprecated, use override_SCTLR_V)
override_MainId	Uns32	Override MIDR register (deprecated, use override_MIDR)
override_CacheType	Uns32	Override CTR register (deprecated, use override_CTR)
override_TLBType	Uns32	Override TLBTR register (deprecated, use override_TLBTR)
override_InstructionAttributes0	Uns32	Override ID_ISAR0 register (deprecated, use override_ISAR0)
override_InstructionAttributes1	Uns32	Override ID_ISAR1 register (deprecated, use override_ISAR1)
override_InstructionAttributes2	Uns32	Override ID_ISAR2 register (deprecated, use override_ISAR2)
override_InstructionAttributes3	Uns32	Override ID_ISAR3 register (deprecated, use override_ISAR3)
override_InstructionAttributes4	Uns32	Override ID_ISAR4 register (deprecated, use override_ISAR4)
override_InstructionAttributes5	Uns32	Override ID_ISAR5 register (deprecated, use override_ISAR5)

Table 8.1: Parameters that can be set in: CPU

8.1 Parameter values and limits

These are the formal parameter limits and actual parameter values

Name	Min	Max	Default	Actual
(Others)				
variant				AArch64
verbose			t	t
suppressCPSWarnings			f	f
showHiddenRegs			f	f
UAL			t	t
enableVFPAAtReset			f	f

SVEImplementedSizes	1	0xffff	15	15
SVEFaultUnknown	0	0xffffffffffff	0xdfdfdfdfdfdfdfdf	0xdfdfdfdfdfdfdfdf
enableSystemBus			f	f
enableSystemMonitorBus			f	f
compatibility			ISA	ISA
unpredictableR15			undefined	undefined
unpredictableModal			undefined	undefined
maxSIMDUnroll	1	16	2	2
override_debugMask	0	0xffffffff	0	0
ASIDCacheSize	0	0x100	8	8
thumbNoCond			f	f
enableHostAtomics			f	f
enableWFEHostAtomics			f	f
endian			none	none
override_fcsePresent			f	f
override_fpexcDexPresent			t	t
override_advSIMDPresent			f	f
override_vfpPresent			f	f
override_physicalBits	32	52	32	32
allowMSRToCurrentBank			f	f
override_SCTLR_V			f	f
override_SCTLR_EE			f	f
override_SCTLR_TE			f	f
override_SCTLR_NMFI			f	f
override_SCTLR_CP15BEN_Present			t	t
override_MIDR	0	0xffffffff	0x410fd000	0x410fd000
override_CTR	0	0xffffffff	0x8404c004	0x8404c004
override_TLBTR	0	0xffffffff	0	0
override_CLIDR	0	0xffffffff	0xa000023	0xa000023
override_AIDR	0	0xffffffff	0	0
override_PFR0	0	0xffffffff	0x131	0x131
override_PFR1	0	0xffffffff	0x11011	0x11011
override_PFR2	0	0xffffffff	0	0
override_DFR0	0	0xffffffff	0x3010066	0x3010066
override_DFR1	0	0xffffffff	0	0
override_AFR0	0	0xffffffff	0	0
override_MMFR0	0	0xffffffff	0x10101105	0x10101105
override_MMFR1	0	0xffffffff	0x40000000	0x40000000
override_MMFR2	0	0xffffffff	0x1260000	0x1260000
override_MMFR3	0	0xffffffff	0x2102211	0x2102211
override_MMFR4	0	0xffffffff	0	0
override_MMFR5	0	0xffffffff	0	0
override_ISAR0	0	0xffffffff	0x2101110	0x2101110
override_ISAR1	0	0xffffffff	0x13112111	0x13112111
override_ISAR2	0	0xffffffff	0x21232042	0x21232042

override_ISAR3	0	0xffffffff	0x1112131	0x1112131
override_ISAR4	0	0xffffffff	0x11142	0x11142
override_ISAR5	0	0xffffffff	1	1
override_ISAR6	0	0xffffffff	0	0
override_PMCR	0	0xffffffff	0x410f3000	0x410f3000
override_PMCEID0	0	0xffffffffffffff	0x3fff0f3f	0x3fff0f3f
override_PMCEID1	0	0xffffffffffffff	0	0
override_PMMIR	0	0xffffffff	0	0
override_DBGDIDR	0	0xffffffff	0x3516c000	0x3516c000
override_DBGDEVID	0	0xffffffff	0x101f13	0x101f13
override_DBGDEVID1	0	0xffffffff	2	2
override_DBGDEVID2	0	0xffffffff	0	0
override_FPSID	0	0xffffffff	0x41033092	0x41033092
override_MVFR0	0	0xffffffff	0x10111222	0x10111222
override_MVFR1	0	0xffffffff	0x12111111	0x12111111
override_MVFR2	0	0xffffffff	67	67
override_FPEXC	0	0xffffffff	0	0
override_ERG	3	11	3	3
override_CCSIDR_1I	0	0xffffffffffffff	0x701fe00a	0x701fe00a
override_CCSIDR_1D	0	0xffffffffffffff	0x201fe00a	0x201fe00a
override_CCSIDR_2I	0	0xffffffffffffff	0x703fe07a	0x703fe07a
override_CCSIDR_2D	0	0xffffffffffffff	0x703fe07a	0x703fe07a
override_CCSIDR_3I	0	0xffffffffffffff	0	0
override_CCSIDR_3D	0	0xffffffffffffff	0	0
override_CCSIDR_4I	0	0xffffffffffffff	0	0
override_CCSIDR_4D	0	0xffffffffffffff	0	0
override_CCSIDR_5I	0	0xffffffffffffff	0	0
override_CCSIDR_5D	0	0xffffffffffffff	0	0
override_CCSIDR_6I	0	0xffffffffffffff	0	0
override_CCSIDR_6D	0	0xffffffffffffff	0	0
override_CCSIDR_7I	0	0xffffffffffffff	0	0
override_CCSIDR_7D	0	0xffffffffffffff	0	0
override_RMR	0	0xffffffff	1	1
override_RVBAR	0	0xffffffffffffff	0	0
override_AA64PFR0_EL1	0	0xffffffffffffff	0x111111	0x111111
override_AA64PFR1_EL1	0	0xffffffffffffff	0	0
override_AA64DFR0_EL1	0	0xffffffffffffff	0x10305106	0x10305106
override_AA64DFR1_EL1	0	0xffffffffffffff	0	0
override_AA64AFR0_EL1	0	0xffffffffffffff	0	0
override_AA64AFR1_EL1	0	0xffffffffffffff	0	0
override_AA64ISAR0_EL1	0	0xffffffffffffff	0	0
override_AA64ISAR1_EL1	0	0xffffffffffffff	0	0
override_AA64ISAR2_EL1	0	0xffffffffffffff	0	0
override_AA64MMFR0_EL1	0	0xffffffffffffff	0x101124	0x101124
override_AA64MMFR1_EL1	0	0xffffffffffffff	0	0

override_AA64MMFR2_EL1	0	0xffffffffffff	0	0
override_DCZID_EL0	0	9	4	4
override_LORID_EL1	0	0xffffffff	0	0
override_STRoffsetPC12			t	t
override_fcseRequiresMMU			f	f
override_ignoreBadCp15			f	f
override_SGIDisable			f	f
override_condUndefined			f	f
override_deviceStrongAligned			t	t
override_stage1SZMinFault			f	f
override_stage1SZMaxFault			f	f
override_stage2SZMinFault			f	f
override_stage2SZMaxFault			f	f
override_mask_ACTLR_EL1	0	0xffffffffffff	0	0
override_mask_ACTLR_EL2	0	0xffffffffffff	115	115
override_mask_ACTLR_EL3	0	0xffffffffffff	115	115
override_Control_V			f	f
override_MainId	0	0xffffffff	0x410fd000	0x410fd000
override_CacheType	0	0xffffffff	0x8404c004	0x8404c004
override_TLBType	0	0xffffffff	0	0
override_InstructionAttributes0	0	0xffffffff	0x2101110	0x2101110
override_InstructionAttributes1	0	0xffffffff	0x13112111	0x13112111
override_InstructionAttributes2	0	0xffffffff	0x21232042	0x21232042
override_InstructionAttributes3	0	0xffffffff	0x1112131	0x1112131
override_InstructionAttributes4	0	0xffffffff	0x11142	0x11142
override_InstructionAttributes5	0	0xffffffff	1	1

Table 8.2: Parameter values and limits

Chapter 9

Execution Modes

Mode	Code
EL0t	0
EL1t	4
EL1h	5
EL2t	8
EL2h	9
EL3t	12
EL3h	13

Table 9.1: Modes implemented in: CPU

Chapter 10

Exceptions

Exception	Code
Reset	0
Undefined	1
SupervisorCall	2
SecureMonitorCall	3
HypervisorCall	4
PrefetchAbort	5
DataAbort	6
HypervisorTrap	7
IRQ	8
FIQ	9
IllegalState	10
MisalignedPC	11
MisalignedSP	12
SError	13

Table 10.1: Exceptions implemented in: CPU

Chapter 11

Hierarchy of the model

A CPU core may be configured to instance many processors of a Symmetrical Multi Processor (SMP). A CPU core may also have sub elements within a processor, for example hardware threading blocks.

OVP processor models can be written to include SMP blocks and to have many levels of hierarchy. Some OVP CPU models may have a fixed hierarchy, and some may be configured by settings in a configuration register. Please see the register definitions of this model.

This model documentation shows the settings and hierarchy of the default settings for this model variant.

11.1 Level 1: CPU

This level in the model hierarchy has 6 commands.

This level in the model hierarchy has 8 register groups:

Group name	Registers
Core_AArch64	33
SIMD_FP_AArch64	32
SVE	49
SVE_artifact	5
AArch64_system	201
AArch64_system_artifact	1
AArch64_SYS_instruction_registers	55
Integration_support	37

Table 11.1: Register groups

This level in the model hierarchy has no children.

Chapter 12

Model Commands

A Processor model can implement one or more **Model Commands** available to be invoked from the simulator command line, from the OP API or from the Imperas Multiprocessor Debugger.

12.1 Level 1: CPU

12.1.1 debugflags

show or modify the processor debug flags

Argument	Type	Description
-get	Boolean	print current processor flags value
-mask	Boolean	print valid debug flag bits
-set	Int32	new processor flags (only flags 0x000003e4 can be modified)

Table 12.1: debugflags command arguments

12.1.2 dumpTLB

report TLB contents

Argument	Type	Description
-all	Boolean	show the contents of all TLBs (if False, show just the current TLB)

Table 12.2: dumpTLB command arguments

12.1.3 isync

specify instruction address range for synchronous execution

Argument	Type	Description
-addresshi	Uns64	end address of synchronous execution range
-addresslo	Uns64	start address of synchronous execution range

Table 12.3: isync command arguments

12.1.4 itrace

enable or disable instruction tracing

Argument	Type	Description
-access	String	show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-after	Uns64	apply after this many instructions
-enable	Boolean	enable instruction tracing
-full	Boolean	turn on all trace features
-instructioncount	Boolean	include the instruction number in each trace
-memory	String	(Alias for access). show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-mode	Boolean	show processor mode changes
-off	Boolean	disable instruction tracing
-on	Boolean	enable instruction tracing
-processorname	Boolean	Include processor name in all trace lines
-registerchange	Boolean	show registers changed by this instruction
-registers	Boolean	show registers after each trace

Table 12.4: itrace command arguments

12.1.5 listSysRegsAA64

12.1.5.1 Argument description

List all AArch64 system registers

12.1.6 validateTLB

check TLB contents against page tables in memory and report incoherent entries

Argument	Type	Description
-all	Boolean	check all TLBs (if False, validate just the current TLB)
-verbose	Boolean	show all TLB entries (if False, show only incoherent entries)

Table 12.5: validateTLB command arguments

Chapter 13

Registers

13.1 Level 1: CPU

13.1.1 Core_AArch64

Registers at level:1, type:CPU group:Core_AArch64

Name	Bits	Initial-Hex	RW	Description
x0	64	0	rw	
x1	64	0	rw	
x2	64	0	rw	
x3	64	0	rw	
x4	64	0	rw	
x5	64	0	rw	
x6	64	0	rw	
x7	64	0	rw	
x8	64	0	rw	
x9	64	0	rw	
x10	64	0	rw	
x11	64	0	rw	
x12	64	0	rw	
x13	64	0	rw	
x14	64	0	rw	
x15	64	0	rw	
x16	64	0	rw	
x17	64	0	rw	
x18	64	0	rw	
x19	64	0	rw	
x20	64	0	rw	
x21	64	0	rw	
x22	64	0	rw	
x23	64	0	rw	
x24	64	0	rw	
x25	64	0	rw	
x26	64	0	rw	
x27	64	0	rw	
x28	64	0	rw	
x29	64	0	rw	frame pointer
x30	64	0	rw	
sp	64	0	rw	stack pointer
pc	64	0	rw	program counter

Table 13.1: Registers at level 1, type:CPU group:Core_AArch64

13.1.2 SIMD_FP_AArch64

Registers at level:1, type:CPU group:SIMD_FP_AArch64

Name	Bits	Initial-Hex	RW	Description
v0	128	-	rw	
v1	128	-	rw	
v2	128	-	rw	
v3	128	-	rw	
v4	128	-	rw	
v5	128	-	rw	
v6	128	-	rw	
v7	128	-	rw	
v8	128	-	rw	
v9	128	-	rw	
v10	128	-	rw	
v11	128	-	rw	
v12	128	-	rw	
v13	128	-	rw	
v14	128	-	rw	
v15	128	-	rw	
v16	128	-	rw	
v17	128	-	rw	
v18	128	-	rw	
v19	128	-	rw	
v20	128	-	rw	
v21	128	-	rw	
v22	128	-	rw	
v23	128	-	rw	
v24	128	-	rw	
v25	128	-	rw	
v26	128	-	rw	
v27	128	-	rw	
v28	128	-	rw	
v29	128	-	rw	
v30	128	-	rw	
v31	128	-	rw	

Table 13.2: Registers at level 1, type:CPU group:SIMD_FP_AArch64

13.1.3 SVE

Registers at level:1, type:CPU group:SVE

Name	Bits	Initial-Hex	RW	Description
z0	512	-	rw	
z1	512	-	rw	
z2	512	-	rw	
z3	512	-	rw	
z4	512	-	rw	
z5	512	-	rw	
z6	512	-	rw	

z7	512	-	rw	
z8	512	-	rw	
z9	512	-	rw	
z10	512	-	rw	
z11	512	-	rw	
z12	512	-	rw	
z13	512	-	rw	
z14	512	-	rw	
z15	512	-	rw	
z16	512	-	rw	
z17	512	-	rw	
z18	512	-	rw	
z19	512	-	rw	
z20	512	-	rw	
z21	512	-	rw	
z22	512	-	rw	
z23	512	-	rw	
z24	512	-	rw	
z25	512	-	rw	
z26	512	-	rw	
z27	512	-	rw	
z28	512	-	rw	
z29	512	-	rw	
z30	512	-	rw	
z31	512	-	rw	
p0	64	0	rw	
p1	64	0	rw	
p2	64	0	rw	
p3	64	0	rw	
p4	64	0	rw	
p5	64	0	rw	
p6	64	0	rw	
p7	64	0	rw	
p8	64	0	rw	
p9	64	0	rw	
p10	64	0	rw	
p11	64	0	rw	
p12	64	0	rw	
p13	64	0	rw	
p14	64	0	rw	
p15	64	0	rw	
FFR	64	0	rw	

Table 13.3: Registers at level 1, type:CPU group:SVE

13.1.4 SVE_artifact

Registers at level:1, type:CPU group:SVE_artifact

Name	Bits	Initial-Hex	RW	Description
VL	64	1	r-	SVE vector length
VLx2	64	2	r-	SVE vector length x 2
VLx4	64	4	r-	SVE vector length x 4
VLx8	64	8	r-	SVE vector length x 8
VLx16	64	10	r-	SVE vector length x 16

Table 13.4: Registers at level 1, type:CPU group:SVE_artifact

13.1.5 AArch64_system

Registers at level:1, type:CPU group:AArch64_system

Name	Bits	Initial-Hex	RW	Description
ACTLR_EL1	64	0	rw	Auxiliary Control (EL1)
ACTLR_EL2	64	0	rw	Auxiliary Control (EL2)
ACTLR_EL3	64	0	rw	Auxiliary Control (EL3)
AFSR0_EL1	32	0	rw	Auxiliary Fault Status 0 (EL1)
AFSR0_EL2	32	0	rw	Auxiliary Fault Status 0 (EL2)
AFSR0_EL3	32	0	rw	Auxiliary Fault Status 0 (EL3)
AFSR1_EL1	32	0	rw	Auxiliary Fault Status 1 (EL1)
AFSR1_EL2	32	0	rw	Auxiliary Fault Status 1 (EL2)
AFSR1_EL3	32	0	rw	Auxiliary Fault Status 1 (EL3)
AIDR_EL1	32	0	r-	Auxiliary ID
AMAIR_EL1	64	0	rw	Auxiliary Memory Attribute Indirection (EL1)
AMAIR_EL2	64	0	rw	Auxiliary Memory Attribute Indirection (EL2)
AMAIR_EL3	64	0	rw	Auxiliary Memory Attribute Indirection (EL3)
CCSIDR_EL1	32	701fe00a	r-	Current Cache Size ID
CLIDR_EL1	32	a000023	r-	Cache Level ID
CNTFRQ_EL0	32	4c4b40	rw	Counter-Timer Frequency
CNTHCTL_EL2	32	3	rw	Counter-Timer Hypervisor Control
CNTHP_CTL_EL2	32	0	rw	Counter-Timer Physical Timer Control (EL2)
CNTHP_CVAL_EL2	64	0	rw	Counter-Timer Physical Timer CompareValue (EL2)
CNTHP_TVAL_EL2	32	0	rw	Counter-Timer Physical Timer TimerValue (EL2)
CNTKCTL_EL1	32	0	rw	Counter-Timer Kernel Control
CNTPCT_EL0	64	0	r-	Counter-Timer Physical Count
CNTPS_CTL_EL1	32	0	rw	Counter-Timer Secure Physical Timer Control
CNTPS_CVAL_EL1	64	0	rw	Counter-Timer Secure Physical Timer CompareValue
CNTPS_TVAL_EL1	32	0	rw	Counter-Timer Secure Physical Timer TimerValue
CNTP_CTL_EL0	32	0	rw	Counter-Timer Physical Timer Control
CNTP_CVAL_EL0	64	0	rw	Counter-Timer Physical Timer CompareValue
CNTP_TVAL_EL0	32	0	rw	Counter-Timer Physical Timer TimerValue
CNTVCT_EL0	64	0	r-	Counter-Timer Virtual Count
CNTVOFF_EL2	64	0	rw	Counter-Timer Virtual Offset
CNTV_CTL_EL0	32	0	rw	Counter-Timer Virtual Timer Control
CNTV_CVAL_EL0	64	0	rw	Counter-Timer Virtual Timer CompareValue
CNTV_TVAL_EL0	32	0	rw	Counter-Timer Virtual Timer TimerValue
CONTEXTIDR_EL1	32	0	rw	Context ID (EL1)
CPACR_EL1	32	0	rw	Architectural Feature Access Control
CPTR_EL2	32	32ff	rw	Architectural Feature Trap (EL2)
CPTR_EL3	32	0	rw	Architectural Feature Trap (EL3)
CSSELR_EL1	32	0	rw	Current Size Selection
CTR_EL0	32	8404c004	r-	Cache Type
CurrentEL	32	c	r-	Current Exception Level
DAIF	32	3c0	rw	Interrupt Mask Bits
DBGAUTHSTATUS_EL1	32	aa	r-	Debug Authentication Status
DBGBCR0_EL1	32	1e0	rw	Debug Breakpoint Control 0
DBGBCR1_EL1	32	1e0	rw	Debug Breakpoint Control 1
DBGBCR2_EL1	32	1e0	rw	Debug Breakpoint Control 2
DBGBCR3_EL1	32	1e0	rw	Debug Breakpoint Control 3
DBGBCR4_EL1	32	1e0	rw	Debug Breakpoint Control 4

DBGBCR5_EL1	32	1e0	rw	Debug Breakpoint Control 5
DBGBVR0_EL1	64	0	rw	Debug Breakpoint Value 0
DBGBVR1_EL1	64	0	rw	Debug Breakpoint Value 1
DBGBVR2_EL1	64	0	rw	Debug Breakpoint Value 2
DBGBVR3_EL1	64	0	rw	Debug Breakpoint Value 3
DBGBVR4_EL1	64	0	rw	Debug Breakpoint Value 4
DBGBVR5_EL1	64	0	rw	Debug Breakpoint Value 5
DBGCLAIMCLR_EL1	32	0	rw	Debug Claim Tag Clear
DBGCLAIMSET_EL1	32	0	rw	Debug Claim Tag Set
DBGDTRTRX_EL0	32	0	rw	Debug Data Transfer, Transmit/Receive
DBGDTR_EL0	64	0	rw	Debug Data Transfer
DBGPRCR_EL1	32	0	rw	Debug Power Control
DBGWCR0_EL1	32	0	rw	Debug Watchpoint Control 0
DBGWCR1_EL1	32	0	rw	Debug Watchpoint Control 1
DBGWCR2_EL1	32	0	rw	Debug Watchpoint Control 2
DBGWCR3_EL1	32	0	rw	Debug Watchpoint Control 3
DBGWVR0_EL1	64	0	rw	Debug Watchpoint Value 0
DBGWVR1_EL1	64	0	rw	Debug Watchpoint Value 1
DBGWVR2_EL1	64	0	rw	Debug Watchpoint Value 2
DBGWVR3_EL1	64	0	rw	Debug Watchpoint Value 3
DCZID_EL0	32	4	r-	Data Cache Zero ID
DLR_EL0	64	0	rw	Debug Link
DSPSR_EL0	32	0	rw	Debug Saved Program Status
ELR_EL1	64	0	rw	Exception Link (EL1)
ELR_EL2	64	0	rw	Exception Link (EL2)
ELR_EL3	64	0	rw	Exception Link (EL3)
ESR_EL1	32	0	rw	Exception Syndrome (EL1)
ESR_EL2	32	0	rw	Exception Syndrome (EL2)
ESR_EL3	32	0	rw	Exception Syndrome (EL3)
FAR_EL1	64	0	rw	Fault Address (EL1)
FAR_EL2	64	0	rw	Fault Address (EL2)
FAR_EL3	64	0	rw	Fault Address (EL3)
FPCR	32	0	rw	Floating Point Control
FPSR	32	0	rw	Floating Point Status
HACR_EL2	32	0	rw	Hypervisor Auxiliary Control
HCR_EL2	64	80000000	rw	Hypervisor Configuration
HPFAR_EL2	64	0	rw	Hypervisor IPA Fault Address
HSTR_EL2	32	0	rw	Hypervisor System Trap
ID_AA64AFR0_EL1	64	0	r-	AArch64 Auxiliary Feature 0
ID_AA64AFR1_EL1	64	0	r-	AArch64 Auxiliary Feature 1
ID_AA64DFR0_EL1	64	10305106	r-	AArch64 Debug Feature 0
ID_AA64DFR1_EL1	64	0	r-	AArch64 Debug Feature 1
ID_AA64ISAR0_EL1	64	0	r-	AArch64 Instruction Set Attribute 0
ID_AA64ISAR1_EL1	64	0	r-	AArch64 Instruction Set Attribute 1
ID_AA64MMFR0_EL1	64	101124	r-	AArch64 Memory Model Feature 0
ID_AA64MMFR1_EL1	64	0	r-	AArch64 Memory Model Feature 1
ID_AA64PFR0_EL1	64	1 00111111	r-	AArch64 Processor Feature 0
ID_AA64PFR1_EL1	64	0	r-	AArch64 Processor Feature 1
ID_AA64ZFR0_EL1	64	0	r-	AArch64 SVE Feature 0
ID_AFR0_EL1	32	0	r-	Auxiliary Feature 0
ID_DFR0_EL1	32	0	r-	Debug Feature 0
ID_ISAR0_EL1	32	0	r-	Instruction Set Attribute 0
ID_ISAR1_EL1	32	0	r-	Instruction Set Attribute 1
ID_ISAR2_EL1	32	0	r-	Instruction Set Attribute 2
ID_ISAR3_EL1	32	0	r-	Instruction Set Attribute 3
ID_ISAR4_EL1	32	0	r-	Instruction Set Attribute 4

ID.ISAR5_EL1	32	0	r-	Instruction Set Attribute 5
ID.MMFR0_EL1	32	0	r-	Memory Model Feature 0
ID.MMFR1_EL1	32	0	r-	Memory Model Feature 1
ID.MMFR2_EL1	32	0	r-	Memory Model Feature 2
ID.MMFR3_EL1	32	0	r-	Memory Model Feature 3
ID.PFR0_EL1	32	0	r-	Processor Feature 0
ID.PFR1_EL1	32	0	r-	Processor Feature 1
ISR_EL1	32	0	r-	Interrupt Status
MAIR_EL1	64	44e048e0 00098aa4	rw	Memory Attribute Indirection (EL1)
MAIR_EL2	64	0	rw	Memory Attribute Indirection (EL2)
MAIR_EL3	64	44e048e0 00098aa4	rw	Memory Attribute Indirection (EL3)
MDCCINT_EL1	32	0	rw	Monitor DCC Interrupt Enable
MDCCSR_EL0	32	0	r-	Monitor DCC Status
MDCR_EL2	32	6	rw	Monitor Debug Configuration (EL2)
MDCR_EL3	32	0	rw	Monitor Debug Configuration (EL3)
MDRAR_EL1	64	0	r-	Monitor Debug ROM Address
MDSCR_EL1	32	0	rw	Monitor Debug System Control
MIDR_EL1	32	410fd000	r-	Main ID
MPIDR_EL1	64	c0000000	r-	Multiprocessor Affinity
MVFR0_EL1	32	0	r-	Media and VFP Feature 0
MVFR1_EL1	32	0	r-	Media and VFP Feature 1
MVFR2_EL1	32	0	r-	Media and VFP Feature 2
NZCV	32	0	rw	Condition Flags
OSDLR_EL1	32	0	rw	OS Double Lock
OSDTRRX_EL1	32	0	rw	OS Lock Data Transfer, Receive
OSDTRTX_EL1	32	0	rw	OS Lock Data Transfer, Transmit
OSECCR_EL1	32	0	rw	OS Lock Exception Catch Control
OSLAR_EL1	32	-	-w	OS Lock Access
OSLSR_EL1	32	a	r-	OS Lock Status
PAR_EL1	64	0	rw	Physical Address
PMCCFILTR_EL0	32	0	rw	Performance Monitors Cycle Count Filter
PMCCNTR_EL0	64	0	rw	Performance Monitors Cycle Count
PMCEID0_EL0	32	3fff0f3f	r-	Performance Monitors Common Event ID 0
PMCEID1_EL0	32	0	r-	Performance Monitors Common Event ID 1
PMCNTENCLR_EL0	32	0	rw	Performance Monitors Count Enable Clear
PMCNTENSET_EL0	32	0	rw	Performance Monitors Count Enable Set
PMCR_EL0	32	410f3040	rw	Performance Monitors Control
PMEVCNTR0_EL0	32	0	rw	Performance Monitors Event Count 0
PMEVCNTR1_EL0	32	0	rw	Performance Monitors Event Count 1
PMEVCNTR2_EL0	32	0	rw	Performance Monitors Event Count 2
PMEVCNTR3_EL0	32	0	rw	Performance Monitors Event Count 3
PMEVCNTR4_EL0	32	0	rw	Performance Monitors Event Count 4
PMEVCNTR5_EL0	32	0	rw	Performance Monitors Event Count 5
PMEVTYPER0_EL0	32	0	rw	Performance Monitors Event Type 0
PMEVTYPER1_EL0	32	0	rw	Performance Monitors Event Type 1
PMEVTYPER2_EL0	32	0	rw	Performance Monitors Event Type 2
PMEVTYPER3_EL0	32	0	rw	Performance Monitors Event Type 3
PMEVTYPER4_EL0	32	0	rw	Performance Monitors Event Type 4
PMEVTYPER5_EL0	32	0	rw	Performance Monitors Event Type 5
PMINTENCLR_EL1	32	0	rw	Performance Monitors Interrupt Enable Clear
PMINTENSET_EL1	32	0	rw	Performance Monitors Interrupt Enable Set
PMOVSCLR_EL0	32	0	rw	Performance Monitors Overflow Flag Status Clear
PMOVSSET_EL0	32	0	rw	Performance Monitors Overflow Flag Status Set
PMSELR_EL0	32	0	rw	Performance Monitors Event Counter Selection

PMSWINC_EL0	32	-	-w	Performance Monitors Software Increment
PMUSERENR_EL0	32	0	rw	Performance Monitors User Enable
PMXEVCNTR_EL0	32	0	rw	Performance Monitors Selected Event Count
PMXEVTYPER_EL0	32	0	rw	Performance Monitors Selected Event Type
REVIDR_EL1	32	0	r-	Revision ID
RVBAR_EL3	64	0	r-	Reset Vector Base Address (EL3)
SCR_EL3	32	430	rw	Secure Configuration
SCTLR_EL1	32	30d50980	rw	System Control (EL1)
SCTLR_EL2	32	30c50830	rw	System Control (EL2)
SCTLR_EL3	32	30c50830	rw	System Control (EL3)
SPSR_EL1	32	0	rw	Saved Program Status (EL1)
SPSR_EL2	32	0	rw	Saved Program Status (EL2)
SPSR_EL3	32	0	rw	Saved Program Status (EL3)
SPSR_abt	32	0	rw	Saved Program Status (Abort Mode)
SPSR_fiq	32	0	rw	Saved Program Status (FIQ Mode)
SPSR_irq	32	0	rw	Saved Program Status (IRQ Mode)
SPSR_und	32	0	rw	Saved Program Status (Undefined Mode)
SPSel	32	1	rw	Stack Pointer Select
SP_EL0	64	0	rw	Stack Pointer (EL0)
SP_EL1	64	0	rw	Stack Pointer (EL1)
SP_EL2	64	0	rw	Stack Pointer (EL2)
SP_EL3	64	0	rw	Stack Pointer (EL3)
TCR_EL1	64	0	rw	Translation Control (EL1)
TCR_EL2	32	80800000	rw	Translation Control (EL2)
TCR_EL3	32	80800000	rw	Translation Control (EL3)
TPIDRRO_EL0	64	0	rw	Thread Pointer/ID, Read-Only (EL0)
TPIDR_EL0	64	0	rw	Thread Pointer/ID (EL0)
TPIDR_EL1	64	0	rw	Thread Pointer/ID (EL1)
TPIDR_EL2	64	0	rw	Thread Pointer/ID (EL2)
TPIDR_EL3	64	0	rw	Thread Pointer/ID (EL3)
TTBR0_EL1	64	0	rw	Translation Table Base 0 (EL1)
TTBR0_EL2	64	0	rw	Translation Table Base 0 (EL2)
TTBR0_EL3	64	0	rw	Translation Table Base 0 (EL3)
TTBR1_EL1	64	0	rw	Translation Table Base 1 (EL1)
VBAR_EL1	64	0	rw	Vector Base Address (EL1)
VBAR_EL2	64	0	rw	Vector Base Address (EL2)
VBAR_EL3	64	0	rw	Vector Base Address (EL3)
VMPIDR_EL2	64	c0000000	rw	Virtualization Multiprocessor ID
VPIDR_EL2	32	410fd000	rw	Virtualization Processor ID
VTCR_EL2	32	80000000	rw	Virtualization Translation Control
VTTBR_EL2	64	0	rw	Virtualization Translation Table Base
ZCR_EL1	32	4	rw	SVE Control (EL1)
ZCR_EL2	32	4	rw	SVE Control (EL2)
ZCR_EL3	32	4	rw	SVE Control (EL3)

Table 13.5: Registers at level 1, type:CPU group:AArch64.system

13.1.6 AArch64.system.artifact

Registers at level:1, type:CPU group:AArch64.system.artifact

Name	Bits	Initial-Hex	RW	Description
cpsr	32	3cd	rw	

Table 13.6: Registers at level 1, type:CPU group:AArch64.system.artifact

13.1.7 AArch64.SYS_instruction_registers

Registers at level:1, type:CPU group:AArch64.SYS_instruction_registers

Name	Bits	Initial-Hex	RW	Description
ATS1E0R	64	-	-w	
ATS1E0W	64	-	-w	
ATS1E1R	64	-	-w	
ATS1E1W	64	-	-w	
ATS1E2R	64	-	-w	
ATS1E2W	64	-	-w	
ATS1E3R	64	-	-w	
ATS1E3W	64	-	-w	
ATS12E0R	64	-	-w	
ATS12E0W	64	-	-w	
ATS12E1R	64	-	-w	
ATS12E1W	64	-	-w	
DCCISW	32	-	-w	
DCCIVAC	64	-	-w	
DCCSW	32	-	-w	
DCCVAC	64	-	-w	
DCCVAU	64	-	-w	
DCISW	32	-	-w	
DCIVAC	64	-	-w	
DCZVA	32	-	-w	
ICIALLU	32	-	-w	
ICIALLUIS	32	-	-w	
ICIVAU	64	-	-w	
TLBIALLE1	64	-	-w	
TLBIALLE1IS	64	-	-w	
TLBIALLE2	64	-	-w	
TLBIALLE2IS	64	-	-w	
TLBIALLE3	64	-	-w	
TLBIALLE3IS	64	-	-w	
TLBIASIDE1	64	-	-w	
TLBIASIDE1IS	64	-	-w	
TLBIIPAS2E1	64	-	-w	
TLBIIPAS2E1IS	64	-	-w	
TLBIIPAS2LE1	64	-	-w	
TLBIIPAS2LE1IS	64	-	-w	
TLBIVAAE1	64	-	-w	
TLBIVAAE1IS	64	-	-w	
TLBIVAALE1	64	-	-w	
TLBIVAALE1IS	64	-	-w	
TLBIVAE1	64	-	-w	
TLBIVAE1IS	64	-	-w	
TLBIVAE2	64	-	-w	
TLBIVAE2IS	64	-	-w	
TLBIVAE3	64	-	-w	
TLBIVAE3IS	64	-	-w	
TLBIVALE1	64	-	-w	
TLBIVALE1IS	64	-	-w	
TLBIVALE2	64	-	-w	
TLBIVALE2IS	64	-	-w	
TLBIVALE3	64	-	-w	
TLBIVALE3IS	64	-	-w	
TLBIVMALE1	64	-	-w	

TLBIVMALLEIIS	64	-	-w	
TLBIVMALLS12E1	64	-	-w	
TLBIVMALLS12E1IS	64	-	-w	

Table 13.7: Registers at level 1, type:CPU group:AArch64_SYS_instruction_registers

13.1.8 Integration support

Registers at level:1, type:CPU group:Integration_support

Name	Bits	Initial-Hex	RW	Description
transactPL	32	3	r-	privilege level of current memory transaction
transactAT	32	0	r-	current memory transaction type: PA=1, VA=0
artifactPAR	64	0	r-	result of address translation for artifact write to ATS1CPR etc
PTWBankSelect	8	0	rw	select PTW bank (0 is stage 1, 1 is stage 2, 2-5 are stage 2 walks initiated by stage 1 level 0-3 entry lookups, respectively)
PTWBankValid	8	0	r-	bitmask of valid banks (0x01 is stage 1, 0x02 is stage 2, 0x04-0x20 are stage 2 walks initiated by stage 1 level 0-3 entry lookups, respectively)
PTWAddressValid	8	0	r-	bitmask of valid bits for each of PTWAddressL0...PTWAddressL3, PTWBase, PTWInput and PTWOutput in current bank
PTWAddressNS	8	0	r-	bitmask of Non-Secure bits for each of PTWAddressL0...PTWAddressL3, PTWBase and PTWOutput in current bank (PTWInput bit is always 0)
PTWValueValid	8	0	r-	bitmask of valid bits for each of PTWValueL0...PTWValueL3 in current bank
PTWAddressL0	64	0	r-	current bank PTW address, level 0
PTWAddressL1	64	0	r-	current bank PTW address, level 1
PTWAddressL2	64	0	r-	current bank PTW address, level 2
PTWAddressL3	64	0	r-	current bank PTW address, level 3
PTWValueL0	64	0	r-	current bank PTW value, level 0
PTWValueL1	64	0	r-	current bank PTW value, level 1
PTWValueL2	64	0	r-	current bank PTW value, level 2
PTWValueL3	64	0	r-	current bank PTW value, level 3
PTWBase	64	0	r-	current bank PTW table base address
PTWInput	64	0	r-	current bank PTW input address
PTWOutput	64	0	r-	current bank PTW output address
PTWPgSize	64	0	r-	current bank PTW page size (Valid only when PTWOutput is valid)
PTWIE_L1S	64	0	-w	perform secure EL1 stage 1 page table walk for fetch, filling PTW query registers
PTWD_E1S	64	0	-w	perform secure EL1 stage 1 page table walk for load/store, filling PTW query registers
PTWIE_L1NS	64	0	-w	perform non-secure EL1 stage 1 page table walk for fetch, filling PTW query registers
PTWD_E1NS	64	0	-w	perform non-secure EL1 stage 1 page table walk for load/store, filling PTW query registers
PTWIE_L2	64	0	-w	perform non-secure EL2 page table walk for fetch, filling PTW query registers
PTWD_E2	64	0	-w	perform non-secure EL2 page table walk for load/store, filling PTW query registers
PTWIS2	64	0	-w	perform non-secure stage 2 page table walk for fetch, filling PTW query registers
PTWD_S2	64	0	-w	perform non-secure stage 2 page table walk for load/store, filling PTW query registers

PTWLEL3	64	0	-w	perform EL3 page table walk for fetch, filling PTW query registers
PTWD_EL3	64	0	-w	perform EL3 page table walk for load/store, filling PTW query registers
PTWI_current	64	0	-w	perform current mode page table walk for fetch, filling PTW query registers
PTWD_current	64	0	-w	perform current mode page table walk for load/store, filling PTW query registers
ResetTLBs	8	0	-w	reset all implemented TLBs to initial state
HaltReason	8	0	r-	bit field indicating halt reason
atomicType	8	0	r-	current atomic instruction type (1:atomic, 2:exclusive)
WakeWFINow	8	0	-w	wake up core from WFI state
ambaResponse	64	0	rw	external AMBA-PV response

Table 13.8: Registers at level 1, type:CPU group:Integration_support